Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 **In this question you must show detailed reasoning.** **[5]**

The line $y = 2x + k$ is a tangent to the curve $y = x^2 - 4x + 7$. Find the value of the constant k and the coordinates of the point where the line touches the curve.

2 A circle has equation $x^2 + y^2 = 8$ and a line has equation $y = x + k$.

(a) Show that the line meets the circle only when $-4 \leq k \leq 4$. **[4]**

(b) Find the coordinates of the point where the line touches the circle when $k = 4$. **[2]**

Question 1

$x^2-4x+7=2x+k \Rightarrow x^2-6x+(7-k)=0$. Tangent: $b^2-4ac=36-4(7-k)=8+4k=0$, so $k=-2$. Then $x^2-6x+9=0$, $x=3$, $y=2(3)-2=4$. The point is $(3,4)$

Question 2

(a) $x^2+(x+k)^2=8 \Rightarrow 2x^2+2kx+k^2-8=0$. For real solutions $b^2-4ac=4k^2-8(k^2-8)=64-4k^2\geq 0$, so $k^2\leq 16$, i.e. $-4\leq k\leq 4$

(b) $k=4$: $2x^2+8x+8=0 \Rightarrow (x+2)^2=0$, so $x=-2$ and $y=-2+4=2$. The point is $(-2,2)$